CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
CHANGA

Third Semester of B. Tech. (Backlog) Examination (CL)
May 2011

CL 205 Building Materials and Construction

Date: 21.05.2011, Saturday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Differentiate the following: [5]
(i) Malleability and Ductility (ii) Impact Strength and Tensile Strength

(b) Classify Truss roof. [2]

Q - 2 (a) Draw a neat cross-section of an exogenous tree and briefly explain its parts. [5]

(b) Differentiate thermo plastics and thermosetting plastics. [5]

(c) Explain briefly various properties of a good brick. [4]

OR

Q - 2 (a) What are the advantages of timber as a construction material? [5]

(b) What is the composition of Glass? Give the classification of Glass. [5]

(c) What are the requirements of a good concrete? [4]

Q - 3 (a) Design a dog-legged stair for 3.0 m floor height. The hall size is 3.0 m x 6.0 m. Assume suitable data and states it clearly. Also prepare the dimensional sketch. [6]

(b) Draw neat sketch of solid ground floor. [5]

(c) Give detailed classification of pitch roof. [3]

OR

Q - 3 (a) Draw neat sketch of typical stair showing all components. [6]

(b) What are basic requirements of a good floor? [5]

(c) Write short note on 'A.C. roof covering'. [3]

SECTION – II

Q - 4 (a) Draw neat sketch of Random rubble built to course stone masonry. [2]

(b) Define plastering. What are the objectives of plastering? [2]

(c) Explain the detailed procedure of painting on wooden surfaces. [3]

Q - 5 (a) Draw elevation and plan of alternate courses of Double Flemish bond for 1 brick thick wall. [5]

(b) Differentiate Junctions and Quoins in connections of brick bond. [2]

(c) Design strip footing for a brick wall 30 cm thick and 3.3 m high above the ground level. The wall carries a superimposed load of 97 kN/m run. The soil has unit weight of 17 kN/m^3, angle of repose of 36^0 and safe bearing capacity is 170 kN/m^2. The footing may have lime concrete base with unit weight of 20 kN/m^3 and modulus of rupture equal to 150 kN/m^2. Assume unit weight of masonry is 19.5 kN/m^3. [7]

OR

Q - 5 (a) Draw elevation and plan of alternate courses of English bond for 1 brick thick wall. [5]

(b) What do you understand by composite masonry? Enumerate various types of composite masonry. [2]

(c) Explain the minimum depth criteria of shallow footing. [3]

(d) Draw sketch of Grillage foundation for steel stanchion. [4]

Q - 6 (a) Draw neat sketch of Battened and Ledged door showing all components. [6]

(b) What are the advantages of steel window? Give the standard sizes of ventilators. [3]

(c) Attempt any **Two**: [3]

(i) Write short note on Box sheeting.

(ii) Explain Raking shore method in detail.

(iii) What do you understand by scaffolding? What are the essential requirements of a good scaffolding?

(d) Write a short note on Vibro-piles. [2]

OR

Q - 6 (a) Write short notes on any **Three**: [9]

(i) Pivoted windows (ii) Corner windows

(iii) Revolving doors (iv) Framed and Panelled doors

(b) Discuss any **Two** methods in detail: [3]

(i) Sheet piling (iii) Steel scaffolding (iii) Pit method of underpinning

(c) Sketch common types of raft foundation. [2]

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